Infania Pimentel

Product Designer and UX · Human Factors Engineer · Interaction Systems

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Executive Summary

Human Factors Engineer and product designer with experience designing wearables, physical-digital products, embedded systems, AI-enabled tools, and interaction systems. Background spans human factors engineering, psychology, computer science, and product design and development.

Core Competencies

Product Design · UX Design · Human Factors Engineering · User Research
Usability Testing · Interaction Design · Product Development · Embedded Systems
AI Product Development · Prototyping · Hardware-Software Integration

Education

TUFTS UNIVERSITY, SCHOOL OF ENGINEERING, Medford, MA — May 2026

Master of Science in Human Factors Engineering — **GPA: 3.8**

Thesis: Designed and built a wearable device using visual signals to help people connect in person, validated through usability and IRB-approved user studies demonstrating measurable usability improvements and increased desire to connect.

UNIVERSITY OF NEW MEXICO, Albuquerque, NM — May 2021

Minor: Computer Science — ***Cum Laude***

Bachelor of Arts in Chemistry; Bachelor of Science in Psychology

Product Design & Interaction Systems

TUFTS UNIVERSITY, SCHOOL OF ENGINEERING, Medford, MA

Product Design Engineer & Human Factors Lead (Master's Thesis) — SociaLIGHT — 2024–May 2026

Developed and evaluated a wearable device that helps people connect in person using social-media-inspired visual signals.

- Prototyped the concept through paper sketches and Figma before building functional hardware end-to-end.
- Switched from BLE/WiFi to NFC to simplify setup, and used visual feedback to communicate system state.
- Designed and ran a two-part study (usability testing and an IRB-approved behavioral study), and people felt a significantly stronger desire to connect and start conversations.
- Refined the design across iterations, cutting setup time by 83% and reaching 100% task completion in testing.

CENTER FOR ENGINEERING EDUCATION AND OUTREACH, Medford, MA

Embedded Product Design Engineer — Componentize Platform — 2024–May 2026

Developed an app-store for engineers to browse and connect hardware to a web interface for live control with an embedded AI Assistant.

- Connected sensors, cameras, and microcontrollers to web apps using BLE, WebSocket, and serial communication.
- Designed and embedded an AI assistant into the platform using JavaScript, PyScript, and OpenAI's API.
- Designed the app store's architecture and visual design, defining consistency and APIs that could scale across a growing library.
- Iterated on the design through ongoing developer feedback and prototype testing in Figma and live browsers.

Product Designer & Embedded Systems Engineer — Smart Playground Initiative — 2024

Designed a screenless toy teaching kids computational thinking through play: a 'magic mailbox' customizes stuffed animals.

- Used signal strength (RSSI) to detect and pair with the stuffed animal placed inside, replacing screen-based menu selection.
- Developed LED-based feedback cues to communicate system state (scanning, selection, pairing) without a screen.
- Built a multi-device system where several embedded boards communicated with each other in real time.
- Tested the toy with kids and used their behavior and feedback to refine how the interaction worked.

Human Factors & Medical Technology

TUFTS UNIVERSITY, ENP 110 — HUMAN FACTORS IN MEDICAL TECHNOLOGY, Medford, MA

Medical Technology Consulting Practicum — Michael Wiklund, Instructor — 2025

Completed medical device HFE projects spanning usability testing, UI requirements, and regulatory-compliant warning/alarm design.

- Completed a four-month anatomy and cadaver lab to ground design decisions in real clinical context.
- Conducted formative usability tests of a glucose meter, producing a full test protocol and design recommendations.
- Designed alarms and warnings for medical devices per IEC 60601-1-8 and ANSI Z535 standards.

Research & Technical Experience

BRAY LABS MACHINE SHOP, Medford, MA

Machine Shop Fabrication Technician 2023–2024

Trained students on additive and subtractive manufacturing and advised on design constraints to improve prototype outcomes.

UNIVERSITY OF NEW MEXICO, NEUROSCIENCES, Albuquerque, NM

Research Technician 2021–2023

Conducted EEG and fNIRS research and developed MATLAB/R algorithms modeling hippocampal dopamine activity.

UNIVERSITY OF MASSACHUSETTS AMHERST, Amherst, MA

Polymer Engineering & Physics Research Assistant Summer 2021

Designed experiments to study programmable polymer mechanics. Built 3D simulations to model self-assembling structures in Python.

Technical Skills

Product Design: Figma, Wireframing, Prototyping, Interaction Design, Onshape, SolidWorks

AI & Software: OpenAI API, Python, JavaScript, PyScript, HTML/CSS, MATLAB, R

Embedded Systems & Hardware: ESP32, Raspberry Pi, BLE, NFC, Sensors, Motors, Cameras, LED Systems

Human Factors & Research: Usability Testing, IRB Study Design, Ergonomic Analysis, Statistical Analysis, Risk Analysis, IEC 62366-1, FDA HFE Guidance

Fabrication: Laser Cutting, Waterjet, Additive Manufacturing, DFM, Milling

Languages: English (Fluent), Spanish (Reading/Writing)